TinyBolus Formulary

Peer-review document

Version 0.0.67 · Updated 2026-07-08

Published by Aether Somnus

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Source: github.com/SGP-Prime/tinybolusformulas
Privacy policy: sgp-prime.github.io/tinybolus-site/privacy.html

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This is a reference tool, not a prescribing system. Verify every value against your hospital protocols, product labelling, and the individual patient in front of you before administration.

About this document

This is the human-readable export of the TinyBolus paediatric reference formulary, organised section-by-section as it appears in the application. Each item is presented with its dose specification, applicable age brackets (where present), route, clinical notes, and numerical references pointing into the bibliography at the end of the document.

How to read each item

Items fall into a small number of shapes that map directly onto how the application computes them:

- **Single dose** (e.g., 10 µg/kg): one factor multiplied by weight, often with a hard maximum cap.
- **Dose range** (e.g., 2–3 mg/kg): a low–high range, also weight-scaled, optionally with a maximum cap on either bound.
- **Age-bracketed**: different dose factors for different age cohorts. Rendered as a small table per item.
- **Age-bracketed fixed value**: not weight-scaled, just a string per age band (e.g., laryngoscope blade size).
- **Computed**: piecewise logic that doesn't reduce to a single formula (ETT sizing, fluid maintenance, LMA / supraglottic airway sizing, vital-sign normal ranges).

Hard floors / contraindications

Some items carry an *age* or *weight* floor below which the formulary explicitly declines to suggest a dose. In the app this renders as `∅` (empty-set glyph) with a "not indicated" explanation. Items with such floors are flagged in red below each row.

References

Every dose / range / age bracket / sizing rule carries one or more citations. References are numbered in encounter order throughout this document, with a full bibliography at the end. The numbers in [brackets] after each item refer to entries in that bibliography.

Source

The formulary itself lives at github.com/SGP-Prime/tinybolusformulas as a single machine-readable JSON file. This document is regenerated from that file on demand by `tool/generate_formulary_review_html.dart`; do not hand-edit it.

Contents

EMERGENCY — PALS

Compression rate

VALUE `100–120/min`

References: [1]

Compression depth

RULE `4 cm <1 y · 5 cm ≥1 y`

not exceeding 6 cm at any age

References: [1]

C:V ratio

VALUE `15:2`

15:2 if PBLS-trained · 30:2 untrained

References: [1]

Defibrillation

DOSE `4 J/kg`

stepwise to 8 J/kg for refractory VF / pulseless VT (6th shock on) · adolescent 120–200 J

References: [1], [2]

Cardioversion (synchronised)

DOSE `1 J/kg`

double per attempt to max 4 J/kg

References: [1]

Adrenaline (cardiac arrest)

IV/IO

DOSE `10 µg/kg (max 1 mg)`

repeat every 3–5 min · max 1 mg/dose

References: [1], [3]

Amiodarone

IV/IO

DOSE `5 mg/kg (max 300 mg)`

after 3rd shock · repeat after 5th · max 150 mg subsequent

References: [1]

Atropine

IV

DOSE `20 µg/kg (max 500 µg)`

References: [1], [4], [5]

Adrenaline (bradycardia)

IV/IO

DOSE `1–2 µg/kg`

unresponsive to oxygenation/ventilation · atropine if vagal

References: [1], [4]

Adenosine (1st dose)

IV

DOSE `0.1 mg/kg (max 6 mg)`

References: [1]

Adenosine (2nd dose)

IV

DOSE `0.3 mg/kg (max 18 mg)`

escalate 0.05–0.1 mg/kg steps to max 0.5 mg/kg

References: [1]

Magnesium sulphate (torsades)

IV/IO

DOSE `50 mg/kg (max 2 g)`

References: [1]

Vital Signs

Heart Rate

≤1 mo	120–170 bpm
2–12 mo	100–160 bpm
1–2 y	98–140 bpm
3–5 y	80–120 bpm
6–11 y	75–118 bpm
≥12 y	60–100 bpm

References: [6]

Systolic BP

≤12 mo	72–104 mmHg
1–2 y	86–106 mmHg
3–5 y	89–112 mmHg
6–9 y	97–115 mmHg
10–11 y	102–120 mmHg
12–15 y	110–131 mmHg
≥16 y	90–140 mmHg

References: [6]

Diastolic BP

≤12 mo	37–56 mmHg
1–2 y	42–63 mmHg
3–5 y	46–72 mmHg
6–9 y	57–76 mmHg
10–11 y	61–80 mmHg
12–15 y	64–83 mmHg
≥16 y	60–90 mmHg

References: [6]

Respiratory Rate

≤1 mo	40–60 /min
2–12 mo	30–53 /min
1–3 y	22–37 /min
4–5 y	20–28 /min
6–12 y	18–25 /min
≥13 y	12–20 /min

References: [6]

Tidal Volume

DOSE `5-7 mL/kg`

per IDEAL body weight — if obese use estimated weight · lower end for neonates

References: [7], [8]

Induction

Propofol

IV

≤1 mo	1–2.5 mg/kg
2–12 mo	3–4 mg/kg
1–6 y	3–3.5 mg/kg
7–12 y	2.5–3 mg/kg
≥13 y	2–2.5 mg/kg

≈ halve after sevoflurane induction

References: [9], [10], [11], [12]

Ketamine

IV

DOSE `1–2 mg/kg`

References: [13], [14]

Ketamine (IM)

IM

DOSE `4–5 mg/kg`

when IV access impractical · onset 3–5 min

References: [13], [14]

Rocuronium

IV

DOSE `0.6–1.2 mg/kg`

upper range for RSI

References: [15], [16]

Succinylcholine (RSI)

IV

≤12 mo	2–3 mg/kg — *raised in infants — larger extracellular volume*
≥1 y	1–2 mg/kg

intubating dose · avoid in MH / hyperkalaemia risk

References: [16], [6], [17]

Fentanyl (induction)

IV

≤1 mo 1–3 µg/kg — *neonate: reduce · inject slowly (rigidity) · watch apnoea*

≥2 mo 1–3 µg/kg

References: [14], [18]

Midazolam

IV

≤6 mo 0.1 mg/kg — *<6 mo: prolonged effect · slow injection · hypotension with opioids*

≥7 mo 0.1 mg/kg

CAP `max 5 mg`

References: [14]

Midazolam (oral premed)

PO

DOSE `0.5 mg/kg (max 20 mg)`

References: [14]

Midazolam (buccal premed)

BUCCAL

DOSE `0.3 mg/kg (max 10 mg)`

if oral refused

References: [14]

Midazolam (IN premed)

IN

DOSE `0.2–0.3 mg/kg (max 10 mg)`

5 mg/mL — divide between nostrils

References: [19], [14]

Ondansetron

IV

DOSE `0.1 mg/kg (max 4 mg)`

Contraindicated <1 months

References: [20], [14], [21]

Dexamethasone (PONV)

IV

DOSE `0.15–0.25 mg/kg (max 4 mg)`

References: [20], [22], [14]

Dexamethasone (anti-inflammatory)

IV

DOSE `0.5 mg/kg (max 8 mg)`

airway oedema / post-extubation stridor

References: [23], [24]

Etomidate

IV

DOSE `0.2–0.3 mg/kg`

for haemodynamic stability

References: [14], [6]

Atracurium

IV

≤12 mo	0.3–0.4 mg/kg
≥1 y	0.4–0.5 mg/kg

References: [14], [6]

Dexmedetomidine (IN premed)

IN

DOSE `1–3 µg/kg (max 200 µg)`

100 µg/mL solution · onset ~30 min

References: [25], [26]

Thiopental

IV

≤1 mo	3–4 mg/kg
2–12 mo	5–8 mg/kg
≥1 y	5–6 mg/kg

then 1 mg/kg PRN · titrate to effect

References: [27]

Painkillers

Paracetamol

IV

≤10 kg 7.5 mg/kg

>10 kg 15 mg/kg

CAP `max 1 g`

interval ≥4 h

References: [28], [14], [29]

Ketorolac

IV

DOSE `0.5 mg/kg (max 30 mg)`

Contraindicated <6 months

References: [30], [14]

Morphine

IV

≤1 mo 0.03–0.05 mg/kg

≥2 mo 0.1 mg/kg

References: [31], [14]

Fentanyl (analgesia)

IV

≤1 mo 0.5–1 µg/kg — *neonate: reduce · slow injection · watch apnoea*

≥2 mo 0.5–1 µg/kg

References: [14], [32], [18]

Metamizole

IV

≤11 mo 5 mg/kg — *reduced < 1 y — immature metabolism*

≥1 y 10–20 mg/kg

CAP `max 1 g`

Contraindicated <3 months

References: [33], [34]

Ibuprofen (IV)

IV

DOSE `10 mg/kg (max 400 mg)`

Contraindicated <3 months

not <5 kg · infuse ≥10 min · PDA use is separate

References: [35], [14], [36], [37], [38]

Ibuprofen (oral)

PO

DOSE `5–10 mg/kg (max 400 mg)`

Contraindicated <3 months

not <5 kg

References: [35], [14], [36]

Tramadol

IV/PO

DOSE `1–2 mg/kg (max 100 mg)`

Contraindicated <12 months

q6h · respiratory depression caution

References: [39], [40]

Diclofenac (PO/PR)

PO/PR

DOSE `0.3–1 mg/kg`

Contraindicated <12 months

q8–12h

References: [41], [40]

Paracetamol (oral)

PO

DOSE `15 mg/kg (max 1 g)`

q4–6h

References: [40]

Emergency Drugs

Adrenaline (cardiac arrest)

IV/IO

DOSE `10 µg/kg (max 1 mg)`

repeat every 3–5 min · max 1 mg/dose

References: [1], [3]

Adrenaline (anaphylaxis IM)

IM

DOSE `10 µg/kg (max 500 µg)`

repeat every 5 min PRN · pens: 150 µg 1–5 y · 300 µg 6–12 y · 500 µg >12 y

References: [1], [4]

Adrenaline (anaphylaxis IV)

IV

DOSE `1 µg/kg`

titrate to response · IM adrenaline is first-line

References: [1]

Atropine

IV

DOSE `20 µg/kg (max 500 µg)`

References: [1], [4], [5]

Ephedrine

IV

≤6 mo 0.6–1.2 mg/kg — *~10× older-child dose — indirect agent, immature sympathetics*

≥7 mo 0.1–0.3 mg/kg

ampoule 30 mg/mL — dilute

References: [42], [14], [6]

Phenylephrine

IV

DOSE `5–20 µg/kg (max 500 µg)`

upper range for tet spells · hypotension: start low, titrate

References: [14], [6]

Lidocaine

IV

DOSE `1 mg/kg (max 100 mg)`

antiarrhythmic — refractory VF / pulseless VT

References: [1], [3]

Glucose 10%

IV/IO

DOSE `2 mL/kg`

hypoglycaemia · recheck glucose after

References: [1], [43]

Calcium gluconate 10%

IV/IO

DOSE `0.5 mL/kg (max 20 mL)`

hypocalcaemia · hyperkalaemia (ECG changes; not in arrest) · CCB overdose · slow IV

References: [1], [43]

Sodium bicarbonate 8.4%

IV/IO

DOSE `1 mmol/kg`

1 mL/kg of 8.4% · TCA toxicity / severe acidosis · NOT for hyperkalaemic arrest

References: [1], [43]

Dantrolene

IV

DOSE `2.5 mg/kg`

reconstitute vials in 60 mL water NOW · repeat 1 mg/kg every 5 min · may exceed 10 mg/kg

References: [44], [45]

Midazolam (buccal, status epilepticus)

BUCCAL

≤11 mo	2.5 mg
12 mo – 4 y	5 mg
5–9 y	7.5 mg
≥10 y	10 mg

Contraindicated <3 months

seizure ≥5 min · 3–6 mo: hospital only · repeat once after 5–10 min

References: [46], [1]

Midazolam (IV, status epilepticus)

IV/IO

DOSE `0.15 mg/kg (max 10 mg)`

repeat once · off-label use

References: [47], [48], [1]

Insulin + glucose (hyperkalaemia)

IV

DOSE `0.1 units/kg (max 10 units)`

WITH glucose 10% 5 mL/kg — never insulin alone · check glucose & K+ q15min

References: [49], [1]

Salbutamol (nebulized, hyperkalaemia)

NEBULIZED

VALUE `2.5–5 mg`

2.5 mg ≤25 kg · 5 mg >25 kg · repeat ≤5× · with insulin–glucose

References: [49], [1]

Fluids and Blood

Fluid Bolus

IV

DOSE `10 mL/kg (max 500 mL)`

balanced crystalloid · cardiogenic shock: 5 mL/kg

References: [1]

Maintenance

RULE `Holliday-Segar 4-2-1: 4 mL/kg/h (first 10 kg) · 2 mL/kg/h (next 10 kg) · 1 mL/kg/h thereafter`

4-2-1 (Holliday-Segar) · isotonic + glucose ± K+ · <1 mo: neonatal regimen

References: [50], [51]

Total Blood Volume

≤1 mo	85 mL/kg
2 mo – 2 y	75 mL/kg
≥3 y	70 mL/kg

References: [6]

Tranexamic Acid

IV

DOSE `15 mg/kg (max 1 g)`

loading dose over 10 min · then 2 mg/kg/h infusion

References: [52], [53], [54]

Max allowable blood loss

VALUE `EBV × (Hct − min Hct) / Hct`

estimate, not a transfusion trigger · EBV ~70–100 mL/kg

References: [55], [56]

Equipment

ETT

RULE `Cuffed: 3.5 + age/4, rounded to ±0.5`

<1 MO `Neonatal weight table (≤4 kg, uncuffed):`

≤1.5 kg	size 2.5
≤3 kg	size 3.0
≤4 kg	size 3.5

INFANT `age table (cuffed):`

≤6 mo	size 3.5
7–12 mo	size 4.0

cuffed sizes — uncuffed add 0.5 (≥1 y) · neonatal table: uncuffed sizes

References: [57], [58]

ETT insertion depth

RULE `6 + weight (<1 mo) · weight/2 + 8 (<1 y) · age/2 + 12 (≥1 y)` *(cm)*

nasal: +1 cm <1 y · +2–3 cm ≥1 y

References: [59], [60], [43]

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iGel LMA

IGEL SIZING `weight bands (inclusive on both bounds):`

2–5 kg	size 1
5–12 kg	size 1.5
10–25 kg	size 2
25–35 kg	size 2.5
30–60 kg	size 3
50–90 kg	size 4
90–∞ kg	size 5

References: [61], [62]

AuraGain LMA

AURAGAIN SIZING `weight bands (inclusive on both bounds):`

2–5 kg	size 1
5–10 kg	size 1.5
10–20 kg	size 2
20–30 kg	size 2.5
30–50 kg	size 3
50–70 kg	size 4
70–100 kg	size 5
100–∞ kg	size 6

References: [63]

Laryngoscope blade

≤1 y	Miller 0–1
2–7 y	Macintosh 2
≥8 y	Macintosh 3

Miller lifts the epiglottis · Macintosh 1 equally effective

References: [64], [6]

Face mask

≤1 mo	Size 0
2–12 mo	Size 1
1–3 y	Size 2
4–8 y	Size 3
9–14 y	Size 4
≥15 y	Size 5

cover nose to chin cleft — anatomical fit over size

References: [6], [65]

Reservoir bag

≤6 mo	0.5 L
7 mo – 6 y	1 L
≥7 y	2 L

References: [66], [67]

Suction catheter

≤1 mo	6 Fr
2 mo – 1 y	8 Fr
2–8 y	10 Fr
≥9 y	12–14 Fr

References: [43], [6]

NG/OG tube

≤1 mo	6 Fr
2 mo – 1 y	8 Fr
2–8 y	10 Fr
≥9 y	12–14 Fr

larger sizes for adolescent decompression

References: [43], [6]

OPA / NPA sizing

VALUE `by landmark`

OPA: incisors → angle of mandible · NPA: nostril → tragus; not in basal-skull fracture

References: [48], [68], [69]

Respiratory

Salbutamol (inhaled)

INHALED

≤5 y 2–10 puffs

≥6 y 4–10 puffs

100 µg/puff · repeat incrementally as needed

References: [70], [71]

Salbutamol (nebulized)

NEBULIZED

≤5 y 2.5 mg

≥6 y 5 mg

References: [70], [71]

Ipratropium (nebulized)

NEBULIZED

≤11 y 250 µg

≥12 y 500 µg

References: [14]

Magnesium Sulphate

IV

DOSE `25–50 mg/kg (max 2 g)`

severe asthma · infuse over 20 min

References: [72], [14], [1]

Clemastine

IV

DOSE `25 µg/kg (max 2 mg)`

Contraindicated <12 months

second-line, after adrenaline

References: [73], [74], [75]

Adrenaline (nebulized)

NEBULIZED

DOSE `0.5 mg/kg (max 5 mg)`

croup / stridor · 1:1000 (1 mg/mL)

References: [76], [14]

Propofol (laryngospasm)

IV

DOSE `0.5 mg/kg`

give in increments

References: [77]

Succinylcholine (laryngospasm)

IV

DOSE `0.1–0.5 mg/kg`

lower end may spare spontaneous ventilation · avoid in MH / hyperkalaemia risk

References: [78], [77]

Succinylcholine (IM)

IM

DOSE `4 mg/kg (max 200 mg)`

laryngospasm / RSI without IV access · avoid in MH / hyperkalaemia risk

References: [78], [77]

Rocuronium (laryngospasm)

IV

DOSE `0.6–1.2 mg/kg`

when succinylcholine contraindicated (MH) — then secure the airway

References: [16], [15]

Salbutamol (IV, severe asthma)

IV

DOSE `15 µg/kg (max 250 µg)`

bolus over 10 min · infusion 1–2 µg/kg/min (>2: PICU) · monitor K+ · off-label <12 y

References: [70], [79]

Hydrocortisone (IV, asthma)

IV

DOSE `4 mg/kg`

q4h — if oral not tolerated

References: [70]

Reversal Agents

Naloxone (titrated)

IV

DOSE `10 µg/kg`

titrate to respiration · repeat q2–3 min

References: [14], [1]

Naloxone (overdose)

IV/IO/IM

DOSE `0.1 mg/kg (max 2 mg)`

References: [1], [14]

Flumazenil

IV

DOSE `10 µg/kg (max 200 µg)`

repeat ≤4× · total max 50 µg/kg (up to 1 mg) · licensed ≥1 y

References: [14]

Sugammadex

IV

DOSE `2–4 mg/kg`

upper dose for deep block

References: [80], [81]

Sugammadex (profound block)

IV

DOSE `16 mg/kg`

CICO rescue — reverses full-dose rocuronium · off-label in children

References: [82], [83]

Neostigmine

IV

DOSE `0.04–0.07 mg/kg (max 5 mg)`

with atropine · slow over 1–2 min, after first recovery signs

References: [14], [6]

Atropine (with neostigmine)

IV

DOSE `20 µg/kg (max 500 µg)`

with or just before neostigmine

References: [14], [6]

Neostigmine + glycopyrronium

IV

DOSE `0.02 mL/kg (max 2 mL)`

References: [84]

Regional Anesthesia

Lidocaine (perineural)

PERINEURAL

≤5 mo maximum 3.5 mg/kg

≥6 mo maximum 5 mg/kg

plain solution — no adrenaline

References: [6], [14]

Bupivacaine (perineural)

PERINEURAL

≤5 mo maximum 1.5–2 mg/kg — *infusion: 0.2 mg/kg/h*

≥6 mo maximum 2–2.5 mg/kg — *infusion: 0.4 mg/kg/h*

References: [85], [86]

Ropivacaine (perineural)

PERINEURAL

≤5 mo maximum 2 mg/kg — *infusion: 0.2 mg/kg/h*

≥6 mo maximum 3 mg/kg — *infusion: 0.4 mg/kg/h*

References: [87], [88], [86]

Intralipid 20% bolus

IV

DOSE `1.5 mL/kg`

LAST rescue · max total 12 mL/kg

References: [85], [89]

Intralipid 20% infusion

IV

DOSE `0.25 mL/kg/min`

LAST rescue – continue until stable

References: [85], [89]

Levobupivacaine (max)

PERINEURAL

DOSE `maximum 1.3 mg/kg (max 150 mg)`

Contraindicated <6 months

References: [90]

Weight Estimation

Used when the user does not supply a measured weight. <1 month and >12 years always require user input.

1–12 mo (infant)

FORMULA `weight (kg) = (0.5 × months) + 4`

References: [43]

1–5 y (young child)

FORMULA `weight (kg) = (2 × age) + 8`

References: [43]

6–12 y (school-age)

FORMULA `weight (kg) = (3 × age) + 7`

References: [91], [43]

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